

Driving Narrator: Real-Time Traffic Sign Detection on Legacy Hardware

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Abstract—This paper demonstrates that modern object detection models can achieve real-time performance on legacy hardware without specialized AI accelerators. Using a 2015 MacBook Air (dual-core Intel i5, no discrete GPU) as a proxy for low-cost automotive ECUs, we deployed a YOLOv11 Nano model trained on 16,000+ augmented traffic sign images. Through INT8 quantization and OpenVINO runtime optimization, the system achieves 14 FPS on 720p video with full processing pipeline—exceeding the 10 FPS threshold for ADAS warning systems—while maintaining 93.2% mAP@0.5 (1.6% degradation from FP32). This independent post-graduation R&D project (June–December 2025) provides a replicable methodology for constraint-first edge AI deployment.

Index Terms—Edge AI, Model Quantization, YOLO, ADAS, OpenVINO, Legacy Hardware

I. INTRODUCTION

The automotive industry faces a deployment gap: sophisticated ADAS systems exist in research, but real-world implementation is constrained by hardware economics. OEMs specify compute platforms years in advance, inheriting architectures spanning 5–7 year roadmaps. The consensus that real-time neural inference requires NPUs/TPUs creates forced upgrade cycles.

For retrofit sectors, millions of vehicles with functional ECUs cannot be upgraded because “the hardware isn’t fast enough.” This project tests a counter-hypothesis: **Can a modern SOTA detector achieve real-time ADAS performance on legacy CPU hardware?**

We target a 2015 MacBook Air (Intel i5-5250U, 8GB RAM, Intel HD 6000—no CUDA) as a proxy for automotive ECUs. Success demonstrates:

- **Retrofit Viability:** Software-only upgrades for existing fleets
- **Cost Reduction:** ADAS on baseline compute without premium hardware
- **Methodology:** Constraint-first optimization as engineering practice

II. METHODOLOGY

A. System Architecture

Hardware: Intel Broadwell i5-5250U (2C/4T, 1.6–2.7GHz), 8GB LPDDR3 shared memory, macOS Monterey.

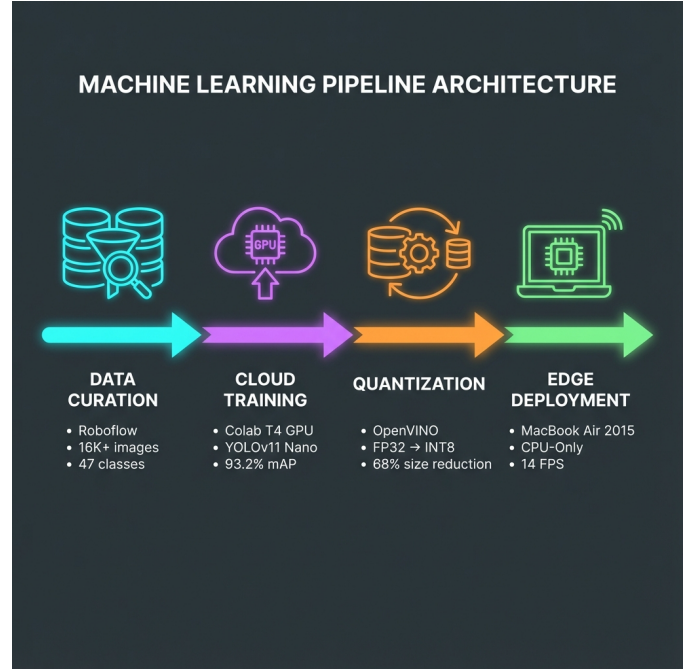


Fig. 1. Pipeline from dataset curation through quantized edge deployment.

B. Dataset

Base: LISA Traffic Sign Dataset [2]—6,600 images, 47 US MUTCD classes.

Augmentation via Roboflow [3]: Brightness $\pm 40\%$, rotation $\pm 15^\circ$, shear $\pm 20\%$, zoom 0.8–1.2 $\times$, 5% cutout occlusion, perspective warp. **Final:** 14,889 train / 993 val / 662 test @ 640 $\times$ 640.

C. Training

Model: YOLOv11 Nano [4]—2.6M parameters, CSPDarknet backbone, PANet neck.

Config: Google Colab T4, batch=32, SGD (lr=0.01 cosine decay), 640 $\times$ 640 train / 416 $\times$ 416 inference. Early stopping patience=10. **Best checkpoint: Epoch 22/25, 93.2% mAP@0.5.**

D. Quantization

Export: PyTorch → ONNX → OpenVINO IR → INT8 (200-image calibration) [5].

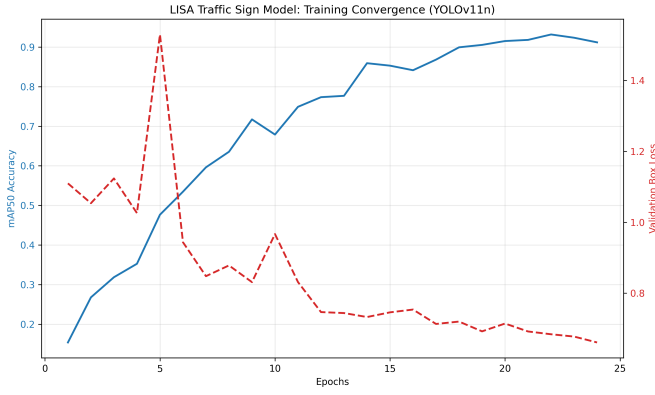


Fig. 2. Training curves showing mAP convergence at epoch 22.

INT8 replaces FP32 with 8-bit integer arithmetic, enabling SIMD vectorization (AVX2). Model size: 10.3MB $\rightarrow$ **3.2MB** (68% reduction).

III. RESULTS

A. Performance

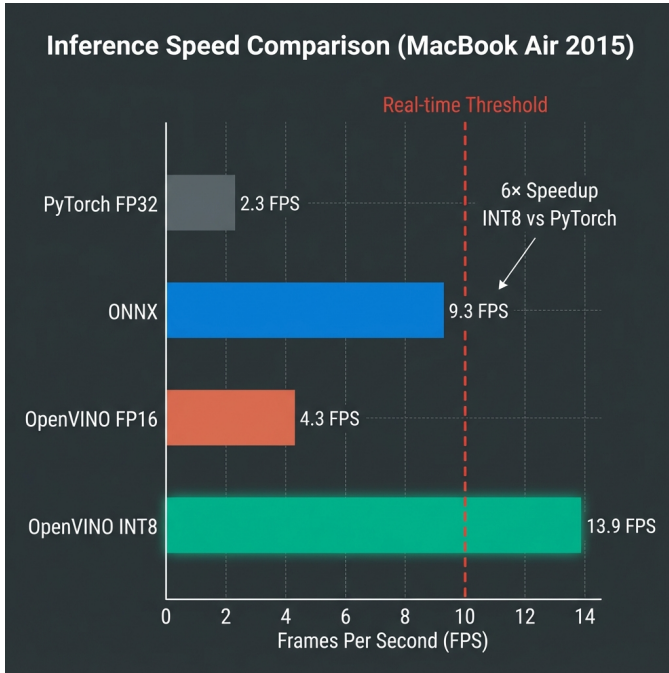


Fig. 3. Inference speed and model size across export formats.

With multithreaded optimization, the full pipeline achieves **14 FPS average** (peaks to 18 FPS), representing a **6 $\times$ speedup** over baseline PyTorch—real-time viable for ADAS warnings.

B. Per-Class Accuracy

High-contrast signs maintain $>90\%$ post-quantization; text-heavy signs show minor degradation.

TABLE I
PERFORMANCE COMPARISON ACROSS MODEL FORMATS

Format	mAP	Raw FPS*	Pipeline FPS [†]	Size	Speedup
PyTorch FP32	94.8%	~ 4	2.3	10.3MB	1.0 $\times$
ONNX	94.8%	~ 8	9.3	10.1MB	4.0 $\times$
OpenVINO FP16	94.8%	~ 4	4.3	5.4MB	1.9 $\times$
OpenVINO INT8	93.2%	~ 24	13.9	3.2MB	6.0$\times$

*Raw inference on static frame after warmup. [†]Multithreaded video pipeline (decode+inference+display) on 720p footage.

TABLE II
TOP PERFORMING SIGN CLASSES

Class	Precision	Recall
Stop	97%	95%
Speed Limit	91%	88%
Yield	89%	92%

C. Failure Modes

Primary errors: False positives on vertical structures (12%), distance-dependent text misreads (8%), severe lighting misses (5%). Conservative 0.25 confidence threshold prioritizes recall—appropriate for warning systems.

IV. FUTURE WORK

- 1) **VPU Acceleration:** While this study focused on CPU constraints, integrating the Intel Neural Compute Stick 2 (NCS2) via OpenVINO could theoretically push inference to 25+ FPS without changing the host hardware.
- 2) **Temporal Consistency:** The current system treats every frame independently. Implementing a lightweight tracker (like ByteTrack) would reduce the flicker of false positives on vertical structures noted in Section III-C.
- 3) **Android Deployment:** The ONNX export pathway validates compatibility with mobile NPU stacks (e.g., Snapdragon SNPE), offering a route to deploy this narrator on dash-mounted smartphones.
- 4) **Audio Feedback Integration:** While this study isolated the computer vision pipeline to benchmark detection latency, the ultimate goal is a “Driving Narrator” system. Preliminary tests with lightweight TTS engines (like pyttsx3) indicate that audio generation adds negligible overhead ($<5\text{ms}$) compared to inference, validating the system’s viability as an auditory ADAS assistant.

V. CONCLUSION

A 10-year-old laptop with no GPU runs a SOTA object detector at real-time speeds when optimized end-to-end. **Core contributions:**

- 1) **Feasibility:** 14 FPS ADAS inference on legacy CPU (>10 FPS threshold)
- 2) **Methodology:** Constraint-first pipeline from data to deployment
- 3) **Transparency:** Documented trade-offs with honest failure analysis



Fig. 4. Detection examples showing successful and edge-case results.

The industry doesn't need next-gen silicon for driver assistance—it needs the right software.

Limitations: Not ISO 26262 certified; US-centric dataset; single-task inference.

Code & Models: <https://github.com/Sasaank79/Driving-Narrator-Legacy-Edge>

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